

UNIVERSITY OF JAMMU

Website: jammuuniversity.ac.in



Practical File on P2CSPC280: Practical (Session: 2025-26)

Submitted By

Danishwer

MCA -2nd Sem

Roll no: 0026MCA25

CERTIFICATE

This is certified to be the Bonafide practical work in C++/HTML/JS in Second Semester of MCA programme during the academic year 2025-26.

This file is prepared by **Danishwer** , roll no. **0026MCA25**, Department of Computer Science and IT.

Teachers-In-Charge:-

1. Prof. Vinod Sharma
2. Dr. Shubhnandan Singh Jamwal
3. Dr. Namrata Karlupia
4. Dr. Rajneet Kour

Internal Examiner

External Examiner

INDEX

DSA

1. Write a program to insert an element at a given position in an array.
2. Write a program to delete an element from a given position in an array.
3. Write a program to search for an element using linear search.
4. Write a program to search for an element using binary search.
5. Write a program to reverse an array (in-place).
6. Write a program to merge two arrays.
7. Write a program to remove duplicate elements from an array.
8. Write a program to implement matrix multiplication.
9. Write a program to create a singly linked list and count the number of nodes.
10. Write a program to insert a node at the beginning, end, and a specific position of a singly linked list.
11. Write a program to reverse a singly linked list.
12. Write a program to merge two singly linked lists.
13. Write a program to implement binary tree traversal (Preorder).

Web Technology:

14. Write a program to enter the marks of students in 4 subjects and find out the sum and average of the marks obtained
15. Write a program to enter a three digit number and find out the sum of all the digits
16. Write a program to enter the number and find if the number is prime or not
17. Write a program to print all the Armstrong numbers from 100 to 500
18. Write a program to print first 20 prime numbers
19. Write a JavaScript program to demonstrate all the events of the Keyboard and Mouse
20. Write a ReactJS Component to find the largest among three numbers
21. Write a NodeJS Code to create a server and send the Rno, Name and Marks in 4 subjects from the client form to the server and return the sum of the marks obtained

Data Structures using C++

1. Write a program to insert an element at a given position in an array.

```
#include <iostream>
using namespace std;

void insert(int item, int index, int *array, int size)
{
    int i = size - 2;
    // Right shifting
    while (i >= index)
    {
        array[i + 1] = array[i];
        i--;
    }
    array[index] = item;
}

int main()
{
    int array[] = {1, 2, 3, 4, 5};
    insert(2, 3, array, 5);
    for (int i : array)
    {
        cout << i << " , ";
    }
    return 0;
}
```

OUTPUT :

1 , 2 , 3 , 2 , 4 ,

2. Write a program to delete an element from a given position in an array.

```
#include <iostream>
using namespace std;
void remove(int index, int *array, int size)
{
    // Left Shifting
    for (int i = index; i < size; i++)
    {
        array[i] = array[i + 1];
    }
    array[size - 1] = 0;
}

int main()
{
    int array[] = {1, 2, 3, 4, 5, 6, 7};
    remove(3, array, 7);
    for (int i : array)
    {
        cout << i << " , ";
    }

    return 0;
}
```

OUTPUT :

1 , 2 , 3 , 5 , 6 , 7 , 0 ,

3. Write a program to search for an element using linear search.

```
#include <iostream>
using namespace std;
int search(int item, int *array, int size)
{
    // Left Shifting
    for (int i = 0; i < size; i++)
    {
        if (array[i] == item)
            return i;
    }
    return -1;
}

int main()
{
    int array[] = {1, 2, 3, 4, 5, 6, 7};

    cout << "Found 7 at : " << search(7, array, 7) << "\n";
    cout << "Found 8 at : " << search(8, array, 7);
    return 0;
}
```

OUTPUT :

```
Found 7 at : 6
Found 8 at : -1
```

4. Write a program to search for an element using binary search.

```
#include <iostream>
using namespace std;
int search(int item, int low, int high, int *array, int
size)
{
    while (low <= high)
    {
        int mid = high - (high - low) / 2; // to avoid
overflow
        int current = array[mid];
        if (item < current)
            return search(item, low, mid - 1, array, size);
        else if (item > current)
            return search(item, mid + 1, high, array, size);
        else
            return mid;
    }
    return -1;
}
int main()
{
    int array[] = {1, 2, 3, 4, 5, 6, 7};
    cout << "Found 2 at : " << search(2, 0, 6, array, 7) <<
"\n";
    cout << "Found 6 at : " << search(6, 0, 6, array, 7) <<
"\n";
    cout << "Found 8 at : " << search(8, 0, 6, array, 7);
    return 0;
}
```

OUTPUT :

```
Found 2 at : 1
Found 6 at : 5
Found 8 at : -1
```

5. Write a program to reverse an array (in-place).

```
#include <iostream>
using namespace std;
void reverse(int *array, int size)
{
    int low = 0;
    int high = size - 1;
    while (low <= high)
    {
        swap(array[low], array[high]);
        low++;
        high--;
    }
}
int main()
{
    int array[] = {1, 2, 3, 4, 5, 6, 7};
    reverse(array, 7);
    for (int i : array)
    {
        cout << i << " , ";
    }
    return 0;
}
```

OUTPUT :

7 , 6 , 5 , 4 , 3 , 2 , 1 ,

6. Write a program to merge two arrays.

```
#include <iostream>
using namespace std;
int *merge(int *array1, int size1, int *array2, int size2)
{
    int total_size = size1 + size2;
    int *result = new int[total_size];
    int i = 0;
    for (; i < size1; i++)
    {
        result[i] = array1[i];
    }
    for (; i < total_size; i++)
    {
        result[i] = array2[i - size1];
    }
    return result;
}
int main()
{
    int array1[] = {1, 2, 3, 4, 5, 6, 7};
    int array2[] = {1, 22, 13, 4};
    int *array3 = merge(array1, 7, array2, 4);
    for (int i = 0; i < 11; i++)
    {
        cout << i[array3] << " , ";
    }
    return 0;
}
```

OUTPUT :

```
1 , 2 , 3 , 4 , 5 , 6 , 7 , 1 , 22 , 13 , 4 ,
```

7. Write a program to remove duplicate elements from an array.

```
#include <iostream>
using namespace std;
void remove_duplicate(int *array, int size)
{
    int *result = new int[size];
    int k = 0;
    for (int i = 0; i < size; i++)
    {
        bool found = false; int current = array[i];
        for (int j = k; j >= 0; j--)
        {
            if (current == result[j])
            {
                found = true; break ;}
        }
        if (!found) {result[k] = current; k++;}
    }
    for (int i = 0; i < k; i++)
    {
        cout << result[i] << " , ";
    }
}
int main()
{
    int array[] = {1, 2, 1, 4, 4, 6, 7, 1, 0, 7, 9, 6};
    for (int i : array)
    {
        cout << i << " , ";
    }
    cout << endl;
    remove_duplicate(array, 12);
    return 0;
}
```

OUTPUT :

```
1 , 2 , 1 , 4 , 4 , 6 , 7 , 1 , 0 , 7 , 9 , 6 ,
1 , 2 , 4 , 6 , 7 , 0 , 9 ,
```

8. Write a program to implement matrix multiplication.

```
#include <iostream>
using namespace std;
void matix_multiplication(int m1[][4], int m2[][2], int r1,
int c1, int r2, int c2)
{
    if (c1 != r2)
    {
        cout << "row in m1 must be equal to col in m2";
        return;
    }
    int **result = new int *[r1];
    for (int i = 0; i < r1; i++)
    {
        result[i] = new int[c2];
    }
    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            int sum = 0;
            for (int k = 0; k < c1; k++)
            {
                sum += m1[i][k] * m2[k][j];
            }
            result[i][j] = sum;
        }
    }
    int r = r1;
    int c = c2;
    cout << "\n";
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            cout << result[i][j] << ",";
        }
    }
}
```

```

        }
        cout << "\n";
    }
    cout << "\n";
}
int main()
{
    int m1[2][4] = {{1, 2, 3, 4}, {5, 6, 7, 8}};
    int m2[4][2] = {{1, 2}, {3, 4}, {5, 6}, {7, 8}};
    for (int i = 0; i < 2; i++)
    {
        for (int j = 0; j < 4; j++)
        {cout << m1[i][j] << ",";
        }cout << "\n";
    }
    cout << "\n";
    for (int i = 0; i < 4; i++)
    {
        for (int j = 0; j < 2; j++)
        {cout << m2[i][j] << ",";
        }cout << "\n";
    }
    matix_multiplication(m1, m2, 2, 4, 4, 2);
    return 0;
}

```

OUTPUT :

```

1,2,3,4,
5,6,7,8,

```

```

1,2,
3,4,
5,6,
7,8,

```

```

50,60,
114,140,

```

9. Write a program to create a singly linked list and count the number of nodes.

```
#include <iostream>
using namespace std;

struct Node
{
    int data;
    Node *node;
};

int count(Node *head)
{
    int number_of_nodes = 0;
    while (head)
    {
        head = head->node;
        number_of_nodes++;
    }
    return number_of_nodes;
}

int main()
{
    Node *n3 = new Node{3, NULL};
    Node *n2 = new Node{3, n3};
    Node *n1 = new Node{3, n2};

    cout << "size : " << count(n1);
    return 0;
}
```

OUTPUT :

```
size : 3
```

10. Write a program to insert a node at the beginning, end, and a specific position of a singly linked list.

```
#include <iostream>
using namespace std;
struct Node
{
    int data;
    Node *next;
};
Node *head;
void print()
{
    Node *temp = head;
    while (temp)
    {
        cout << temp->data << " -->";
        temp = temp->next;
    }
}
void insert_beginning(int value)
{
    Node *new_node = new Node{value, NULL};
    new_node->next = head;
    head = new_node;
}
void insert_end(int value)
{
    Node *new_node = new Node{value, NULL};
    if (!head)
    {
        head = new_node;
        return;
    }
    Node *temp = head;
    while (temp->next)
    {
```

```

        temp = temp->next;
    }
    temp->next = new_node;
}
void insert_location(int value, int index)
{
    if (!head || index == 0)
    {
        insert_beginning(value);
        return;
    }

    Node *new_node = new Node{value, NULL};
    Node *temp = head;
    for (int i = 0; i < index - 1 && temp->next; i++)
    {
        temp = temp->next;
    }
    new_node->next = temp->next;
    temp->next = new_node;
}
int main()
{
    insert_beginning(2);
    insert_beginning(1);
    insert_end(3);
    insert_end(4);
    insert_location(-1, 0);
    insert_location(5, 2);
    print();
    return 0;
}

```

OUTPUT :

```

-1 -->1 -->5 -->2 -->3 -->4 -->
... ..

```

11. Write a program to reverse a singly linked list.

```
#include <iostream>
using namespace std;
struct Node
{
    int data;
    Node *next;
};
Node *head;

void print()
{
    Node *temp = head;
    while (temp)
    {
        cout << temp->data << " -->";
        temp = temp->next;
    }
    cout << "\n";
}

void reverse()
{
    Node *prev = NULL, *next = NULL, *cur = head;
    while (cur)
    {
        next = cur->next;
        cur->next = prev;
        prev = cur;
        cur = next;
    }
    head = prev;
}

int main()
{
```

```
Node *n3 = new Node{3, NULL};
Node *n2 = new Node{2, n3};
head = new Node{1, n2};
print();
reverse();
print();
return 0;
}
```

OUTPUT :

```
1 -->2 -->3 -->
3 -->2 -->1 -->
```

c:\Users\danny\OneDrive\Desktop'

12. Write a program to merge two singly linked lists.

```
#include <iostream>
using namespace std;
#define N(x, y) \
    new Node { x, y }
struct Node
{
    int data;
    Node *next;
};

void print(Node *head)
{
    Node *temp = head;
    while (temp)
    {
        cout << temp->data << " -->";
        temp = temp->next;
    }
    cout << "\n";
}

Node *merge(Node *l1, Node *l2)
{
    if (!l1)
        return l2;
    Node *temp = l1;
    while (temp->next)
    {
        temp = temp->next;
    }
    temp->next = l2;
    return l1;
}

int main()
{
    Node *l1, *l2;
    l1 = N(1, N(2, N(3, NULL)));
```

```
    12 = N(4, N(5, N(6, NULL)));  
    print(l1);  
    print(l2);  
    merge(l1, l2);  
    print(l1);  
    return 0;  
}
```

OUTPUT :

```
1 -->2 -->3 -->  
4 -->5 -->6 -->  
1 -->2 -->3 -->4 -->5 -->6 -->
```

13. Write a program to implement binary tree traversal (Preorder).

```
#include <iostream>
#include <stack>
using namespace std;
struct Node
{
    int data;
    Node *left = NULL, *right = NULL;
};
void printNode(Node *node, int indentation)
{
    if (!node)
        return;

    string indent(indentation, ' ');
    cout << indent << "->" << node->data << "\n";

    printNode(node->left, indentation + 2);
    printNode(node->right, indentation + 2);
}
void printTree(Node *root)
{
    cout << "Tree\n";
    printNode(root, 0);
}

void print(Node *node)
{
    cout << "[" << node->data << "]";
}
void preOrder(Node *root)
{
    if (root == NULL)
        return;
    stack<Node *> s;
    Node *cur = root;

    while (cur)
    {
        print(cur);
        if (cur->right)
            s.push(cur->right);
        if (cur->left)
            cur = cur->left;
        else
```

```

        {
            cur = s.top();
            s.pop();
        }
    }
}

int main()
{
    Node *n5 = new Node{5};
    Node *n4 = new Node{4};
    Node *n3 = new Node{3};
    Node *n2 = new Node{2};
    Node *n1 = new Node{1};
    Node *root = new Node{0};

    root->left = n1;
    root->right = n2;
    n1->left = n3;
    n1->right = n4;
    n2->left = n5;

    printTree(root);
    cout << "Preorder Traversal : ";
    preOrder(root);
    return 0;
}

```

OUTPUT :

```

Tree
->0
  ->1
    ->3
    ->4
  ->2
    ->5
Preorder Traversal : [0][1][3][4][2][5]

```

Web Technology

Index

S.No	Practical Name
1	Write a program to enter the marks of students in 4 subjects and find out the sum and average of the marks obtained
2	Write a program to enter a three digit number and find out the sum of all the digits
3	Write a program to enter the number and find if the number is prime or not
4	Write a program to print all the Armstrong numbers from 100 to 500
5	Write a program to print first 20 prime numbers
6	Write a JavaScript program to demonstrate all the events of the Keyboard and Mouse
7	Write a ReactJS Component to find the largest among three numbers
8	Write a NodeJS Code to create a server and send the Rno, Name and Marks in 4 subjects from the client form to the server and return the sum of the marks obtained

Practical 1 : Write a program to enter the marks of students in 4 subjects and find out the sum and average of the marks obtained

Source Code:

```
<html>

<head>
    <title>Practical 1 - Sum & Average of Marks</title>
</head>

<body>
    <h2>Sum & Average of Marks</h2>
    <input id="s1" type="number" placeholder="Subject 1"><br><br>
    <input id="s2" type="number" placeholder="Subject 2"><br><br>
    <input id="s3" type="number" placeholder="Subject 3"><br><br>
    <input id="s4" type="number" placeholder="Subject 4"><br><br>
    <button onclick="calc()">Calculate</button>
    <p id="res"></p>
    <script>
        function calc() {
            var a = +document.getElementById('s1').value, b =
+document.getElementById('s2').value, c =
+document.getElementById('s3').value, d =
+document.getElementById('s4').value;
            var sum = a + b + c + d;
            document.getElementById('res').innerHTML = "Sum: " + sum +
"<br>Average: " + (sum / 4);
        }
    </script>
</body>

</html>
```

Output:

Sum & Average of Marks	
85	
90	
78	
92	
Calculate	
Sum: 345	
Average: 86.25	

Practical 2 : Write a program to enter a three digit number and find out the sum of all the digits

Source Code:

```
<html>
<head><title>Practical 2 - Sum of Digits</title></head>
<body>
<h2>Sum of Digits of a Three Digit Number</h2>
<input id="num" type="number" placeholder="Enter 3 digit number">
<button onclick="calc()">Find Sum</button>
<p id="res"></p>
<script>
function calc(){
var n=document.getElementById('num').value;
var sum=0;
for(var i=0;i<n.length;i++) sum+=+n[i];
document.getElementById('res').innerText="Sum of digits: "+sum;
}
</script>
</body>
</html>
```

Output:

Sum of Digits of a Three Digit Number	
456	Find Sum
Sum of digits: 15	

Practical 3 : Write a program to enter the number and find if the number is prime or not

Source Code:

```
<html>
<head><title>Practical 3 - Prime Check</title></head>
<body>
<h2>Prime Number Check</h2>
<input id="num" type="number" placeholder="Enter a number">
<button onclick="check()">Check</button>
<p id="res"></p>
<script>
function check(){
var n=+document.getElementById('num').value,prime=n>1;
for(var i=2;i<=Math.sqrt(n);i++){if(n%i==0){prime=false;break;}}
document.getElementById('res').innerText=n+(prime?" is Prime":" is Not
Prime");
}
</script>
</body>
</html>
```

Output:

Prime Number Check

29 is Prime

Practical 4 : Write a program to print all the Armstrong numbers from 100 to 500

Source Code:

```
<html>
<head><title>Practical 4 - Armstrong Numbers</title></head>
<body>
<h2>Armstrong Numbers from 100 to 500</h2>
<button onclick="find()">Find</button>
<p id="res"></p>
<script>
function find(){
var r=[];
for(var i=100;i<=500;i++){
var s=0,t=i.toString();
for(var j=0;j<t.length;j++) s+=Math.pow(+t[j],t.length);
if(s==i) r.push(i);
}
document.getElementById('res').innerText="Armstrong Numbers: "+r.join(",");
}
</script>
</body>
</html>
```

Output:

Armstrong Numbers from 100 to 500

Find

Armstrong Numbers: 153, 370, 371, 407

Practical 5 : Write a program to print first 20 prime numbers

Source Code:

```
<html>
<head><title>Practical 5 - First 20 Primes</title></head>
<body>
<h2>First 20 Prime Numbers</h2>
<button onclick="find()">Find</button>
<p id="res"></p>
<script>
function find(){
var primes=[],n=2;
while(primes.length<20){
var ok=true;
for(var i=2;i<=Math.sqrt(n);i++){if(n%i==0){ok=false;break;}}
if(ok) primes.push(n);
n++;
}
document.getElementById('res').innerText=primes.join(", ");
}
</script>
</body>
</html>
```

Output:

First 20 Prime Numbers

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71

Practical 6 : Write a JavaScript program to demonstrate all the events of the Keyboard and Mouse

Source Code:

```
<html>
<head><title>Practical 6 - Keyboard & Mouse Events</title>
<style>
#box{width:200px;height:200px;background:#ccc;display:flex;align-
items:center;justify-content:center;margin:10px 0;cursor:pointer}
</style>
</head>
<body>
<h2>Keyboard & Mouse Events</h2>
<input id="inp" placeholder="Type here for keyboard events"><br><br>
<div id="box">Hover/Click Me</div>
<p id="log"></p>
<script>
var log=document.getElementById('log');
function show(e){log.innerHTML=e+"<br>" +log.innerHTML;}
var inp=document.getElementById('inp');
inp.onkeydown=function(e){show("keydown: "+e.key);};
inp.onkeyup=function(e){show("keyup: "+e.key);};
inp.onkeypress=function(e){show("keypress: "+e.key);};
var box=document.getElementById('box');
box.onclick=function(){show("click");};
box.ondblclick=function(){show("dblclick");};
box.onmousedown=function(){show("mousedown");};
box.onmouseup=function(){show("mouseup");};
box.onmouseover=function(){show("mouseover");};
box.onmouseout=function(){show("mouseout");};
box.onmousemove=function(e){show("mousemove: "+e.clientX+", "+e.clientY);};
box.oncontextmenu=function(e){e.preventDefault();show("contextmenu (right-
click)");};
</script>
</body>
</html>
```

Output:

Keyboard & Mouse Events

Hello

Hover/Click Me

click

mouseup

mousedown

mousemove: 108,203

mouseover

keyup: o

keypress: o

keydown: o

keyup: l

keypress: l

keydown: l

keyup: l

keypress: l

keydown: l

keyup: e

keypress: e

Practical 7 : Write a ReactJS Component to find the largest among three numbers

Source Code:

```
<html>
<head>
<title>Practical 7 - Largest Among Three (ReactJS)</title>
<script src="https://unpkg.com/react@18/umd/react.development.js"></script>
<script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
<script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
</head>
<body>
<div id="root"></div>
<script type="text/babel">
function App() {
  const [a, setA]=React.useState('');
  const [b, setB]=React.useState('');
  const [c, setC]=React.useState('');
  const [res, setRes]=React.useState('');
  function find() {
    setRes("Largest: "+Math.max(+a, +b, +c));
  }return(<div>
<h2>Find Largest Among Three Numbers (ReactJS)</h2>
<input type="number" placeholder="Number 1" value={a}
onChange={e=>setA(e.target.value)}/><br/><br/>
<input type="number" placeholder="Number 2" value={b}
onChange={e=>setB(e.target.value)}/><br/><br/>
<input type="number" placeholder="Number 3" value={c}
onChange={e=>setC(e.target.value)}/><br/><br/>
<button onClick={find}>Find Largest</button>
<p>{res}</p>
</div>
);
}
ReactDOM.createRoot(document.getElementById('root')).render(<App/>);
</script>
</body>
</html>
```

Output:

Find Largest Among Three Numbers (ReactJS)

Largest: 78

Practical 8 : Write a NodeJS Code to create a server and send the Rno, Name and Marks in 4 subjects from the client form to the server and return the sum of the marks obtained

Source Code (practical8_client.html):

```
<html>

<head>
  <title>Practical 8 - Student Marks Form</title>
</head>

<body>
  <h2>Student Marks Form</h2>
  <form method="POST" action="/">
    <input name="rno" placeholder="Roll No"><br><br>
    <input name="name" placeholder="Name"><br><br>
    <input name="m1" type="number" placeholder="Subject 1"><br><br>
    <input name="m2" type="number" placeholder="Subject 2"><br><br>
    <input name="m3" type="number" placeholder="Subject 3"><br><br>
    <input name="m4" type="number" placeholder="Subject 4"><br><br>
    <button type="submit">Submit</button>
  </form>
</body>

</html>
```

Source Code (practical8_server.js):

```
const http = require('http');
const qs = require('querystring');

http.createServer((req, res) => {
  if (req.method === 'GET') {
    res.end("Server is running");
  } else {
    let body = '';
    req.on('data', d => body += d);
    req.on('end', () => {
      const p = qs.parse(body);
      const sum = +p.m1 + +p.m2 + +p.m3 + +p.m4;
      res.writeHead(200, { 'Content-Type': 'text/html' });
      res.end('<html><body><h2>Result</h2><p>Roll No: ' + p.rno +
        '</p><p>Name: ' + p.name + '</p><p>Marks: ' + p.m1 + ', ' + p.m2 + ', ' +
        p.m3 + ', ' + p.m4 + '</p><p><b>Sum: ' + sum + '</b></p><a
        href="/">Back</a></body></html>');
    });
  }
}).listen(3000, () => console.log('http://localhost:3000'));
```

Output (Before Submit):

Student Marks Form

Output (After Submit):

Result

Roll No: 101

Name: John

Marks: 85, 90, 78, 92

Sum: 345

[Back](#)